

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<think> First let's calculate the distance John traveled before turning around . He drove for 3 hours at a speed of 60 mph . By using the formula $\text{distance} = \text{speed} * \text{time}$ we get $60 * 3 = 180$ miles . Now let's calculate the distance he traveled during his attempt to get home . He spent the first 2 hours in stand still traffic . During these 2 hours he didn't travel any distance . Then he spent the next 0.5 hours driving at a speed of 30 mph . Again using the formula $\text{distance} = \text{speed} * \text{time}$ we get $30 * 0.5 = 15$ miles . Finally he spent the remaining time driving at a speed of 80 mph . To calculate the remaining time we subtract the time he spent in stand still traffic and the time he spent driving at 30 mph from the total time . $4 - 2 - 0.5 = 1.5$ hours . Now using the formula $\text{distance} = \text{speed} * \text{time}$ we get $80 * 1.5 = 120$ miles . Now let's calculate the total distance he traveled trying to get home . $15 + 120 = 135$ miles . Now let's calculate how far he is from home . Initially he traveled 180 miles away from home . Then he traveled 135 miles back towards home . Therefore the distance he is from home is $180 - 135 = 45$ miles . </think><answer>45</answer>

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